Active Management of IO Beta

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April 4, 2016

IO Beta Allocation Strategy: March 2016 Commentary

- **IO Index Returns:** 1.33% for Index with Curve (EDFs+Tsy Futures) and TBA Hedges (March 2016)
- **Fund Beta Allocation YTD:** Outperformed IO Index (w/ Curve Hedges) by 48bps

IO Beta Allocation Strategy: April 2016 Outlook

- **Fixed Income Fund Allocation:** 36% (March: 32%)
- **Model 1 (IO Px/Cpn not duration-adjusted):** 21% (March: 36%)
- **Model 2 (IO Px/Cpn duration-adjusted):** 7% (March: 33%)

Active Management of IO Beta

Investigate if we can design an **active** strategy for investing in IO Beta that outperforms a **passive** strategy that is always 100% invested.

Metrics and Set-up

- Create a **Hedged IO** index
  - Hedge various IOS with current coupon TBA using model durations (Barclays) until December 2014. From 2015 onwards, use YB durations and employ TBA and Curve hedges.
  - Index is created by using monthly market value weights for each IOS based on the market weight of the corresponding cohort
  - Ginnie Mae IOS are excluded
- Simulating Passive IO Beta: Total Return and Sharpe Ratio of Hedged IO index from March 2011 - March 2016
- Simulating Active IO Beta: Dynamic allocation to Hedged IO index based on relative value metrics

Performance of Passive IO Beta Strategy: Mar 2011 - March 2016

- **CAGR**: 6.7%
- **Ann. Volatility**: 15.0%
- **Sharpe Ratio**: 0.5
- **% Positive Months**: 57
- **Best Month**: 11.6%
- **Worst Month**: -12.9%
- **Max Drawdown**: -23.5%
- **Drawdown Period**: June 2011 - March 2013

Framework for Active IO Beta Strategy

- Regress Hedged IO Index returns against RV metrics (1m lag):
  - IO Index Price/Coupon[1]
  - Projected 1-month Carry[2]
  - Convexity: 60-day moving average of realized volatility on the current coupon
  - Prepayment Risk: Incentive for the Mortgage Universe
  - Yield Curve Slope: 10s-2s slope
- Calculate z-score of projected return and allocate to Beta (Net IO) based on score

[1] While OAS would also be useful, in practice dealer model updates are not always appropriately back-filled which makes it challenging to get a consistent time series of values.

[2] This is not available on a historical basis so we use realized carry as a proxy.

Active IO Beta Strategy: Regression Results

- Significant Variables
  - High: IO Index Price/Coupon, Prepayment Risk
  - Medium: Realized Volatility
  - Low: Realized Carry, Yield Curve Slope, Index OAS
- Final Model
  - Proj. Return $\sim$ Px/Cpn + Cnvx + PrepayRisk + Carry

Active IO Beta Strategy: Beta Allocation

- Solve for $k = a + bz$ (z = z-score) such that we optimize the Sharpe Ratio of the strategy $k \times$ Hedged IO Return.
- Set IO $\text{Beta}_{\text{max}} = 100\%$ and IO $\text{Beta}_{\text{min}} = -0.25\%$
- Rescale k so that it is (more or less) mapped to these boundaries.

Active IO Beta Strategy: Two Beta Allocation Models

- Model 1: Don't adjust IO Index Px/Cpn for Interest Rates. $R^2 = 0.33$.
- Model 2: Adjust IO Index Px/Cpn for Interest Rates.[3] $R^2 = 0.17$.

[3] We estimate that the Px/Cpn ratio moves by about 0.14 for each 10bps move in the current coupon.

Active IO Beta Strategy: Model 1 Return Profile

- **CAGR**: 8.9%
- **Ann. Volatility**: 5.4%
- **Sharpe Ratio**: 1.6
- **% Positive Months**: 75
- **Best Month**: 9.3
- **Worst Month**: -0.7%
- **Max Drawdown**: -0.9%
- **Drawdown Period**: November 2011 - January 2012

Active IO Beta Strategy: Model 2 Return Profile

- **CAGR**: 6.5%
- **Ann. Volatility**: 5.2%
- **Sharpe Ratio**: 1.2
- **% Positive Months**: 68
- **Best Month**: 9.1
- **Worst Month**: -2.7%
- **Max Drawdown**: -3.4%
- **Drawdown Period**: September 2011 - January 2012

Track Record of IO Beta Allocation Strategies

	IO Beta Allocation										
	Fund	Model1	Model2	100% Beta CrvAdj	Fund	Model1	Model2	100% Beta CrvAdj	Fund	Model1	Model2
Jan-15	26%	-11%	-16%	-8.9%	-2.31%	0.98%	1.40%				
Feb-15	25%	26%	59%	5.5%	1.38%	1.44%	3.24%				
Mar-15	21%	-7%	-51%	1.2%	0.25%	-0.08%	-0.61%				
Apr-15	15%	-21%	-16%	1.2%	0.18%	-0.26%	-0.19%				
May-15	22%	-28%	-36%	-1.7%	-0.37%	0.47%	0.61%				
Jun-15	31%	-17%	-24%	-0.7%	-0.20%	0.11%	0.16%				
Jul-15	33%	-25%	-51%	1.2%	0.40%	-0.30%	-0.61%				
Aug-15	28%	-42%	-16%	-1.5%	-0.42%	0.64%	0.23%				
Sep-15	24%	-24%	-23%	-1.6%	-0.38%	0.39%	0.37%				
Oct-15	26%	-10%	5%	-0.1%	-0.03%	0.01%	-0.01%				
Nov-15	21%	2%	-5%	1.8%	0.38%	0.04%	-0.08%				
Dec-15	22%	15%	0%	1.4%	0.30%	0.21%	0.01%	**-2.23%**	**-0.85%**	**3.66%**	**4.53%**
Jan-16	17%	12%	3%	-2.1%	-0.36%	-0.25%	-0.07%				
Feb-16	31%	17%	54%	0.5%	0.16%	0.09%	0.28%				
Mar-16	32%	36%	33%	1.3%	0.43%	0.48%	0.44%	**-0.25%**	**0.23%**	**0.32%**	**0.65%**
Apr-16	36%	21%	7%								

As of: Apr 4, 2016

Evolution of IO Beta Allocation RV Variables

Date	MthlyHedgedReturn	OAS	Carry	RealizedVol	YCSlope	PrepayRiskPrem	PxCpnMult	LIB1m	RefiPctPool	RefiPctLoan	RefiIndex	CC	CCChanges
1/30/2015	-8.9	65	1.1	60	121	103	4.78	0.17	0.66	0.66	2585	2.45	-38
2/27/2015	5.5	-87	1.6	73	138	89	5.51	0.17	0.61	0.56	1740	2.74	28
3/31/2015	1.2	-97	1.0	78	138	92	5.37	0.18	0.64	0.62	1942	2.66	-8
4/30/2015	1.2	28	0.2	69	147	81	5.56	0.18	0.64	0.61	1559	2.75	9
5/29/2015	-1.7	27	0.4	68	149	73	5.66	0.18	0.55	0.50	1234	2.81	6
6/30/2015	-0.7	10	0.3	75	170	51	6.16	0.19	0.45	0.40	1246	3.07	26
7/31/2015	1.2	14	0.2	82	153	66	5.93	0.19	0.50	0.42	1418	2.92	-16
8/31/2015	-1.5	21	0.5	76	146	68	5.92	0.20	0.55	0.53	1546	2.93	2
9/30/2015	-1.6	40	0.6	67	142	71	5.70	0.19	0.52	0.49	1590	2.80	-13
10/30/2015	-0.1	42	0.6	66	142	68	5.69	0.19	0.58	0.55	1535	2.84	4
11/30/2015	1.8	3	0.6	58	129	57	5.89	0.24	0.48	0.45	1489	2.95	10
12/31/2015	1.4	5	0.6	61	122	53	5.97	0.43	0.45	0.41	1667	3.01	6
1/29/2016	-2.1	21	0.7	57	116	74	5.38	0.43	0.62	0.60	1734	2.65	-36
2/29/2016	0.5	-7	1.6	67	95	92	5.07	0.44	0.67	0.63	2100	2.53	-12
3/31/2016	1.3	-6	1.4	65	105	93	5.16	0.44	0.67	0.63	1672	2.54	1